

Logistic Regression

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November 5, 2024

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Today, we'll learn...

- What is logistic regression used for?
- How do logistic regression models work?
- How can we use logistic regression for classification?

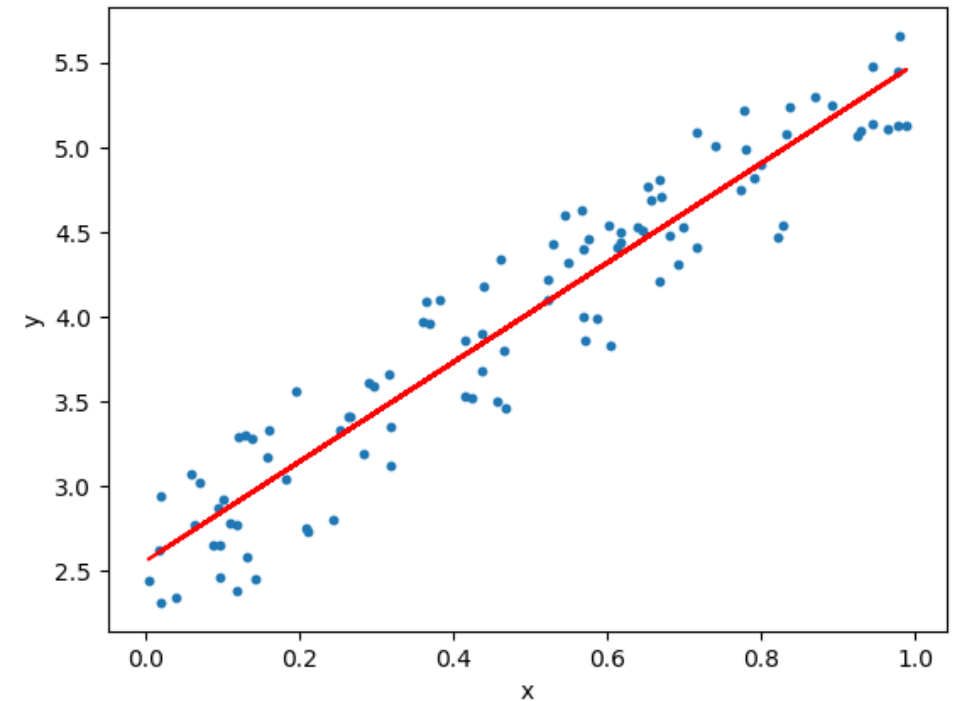
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Review of simple linear regression

In linear regression, we want to predict a *continuous* outcome by fitting a linear model to the data.

Example:



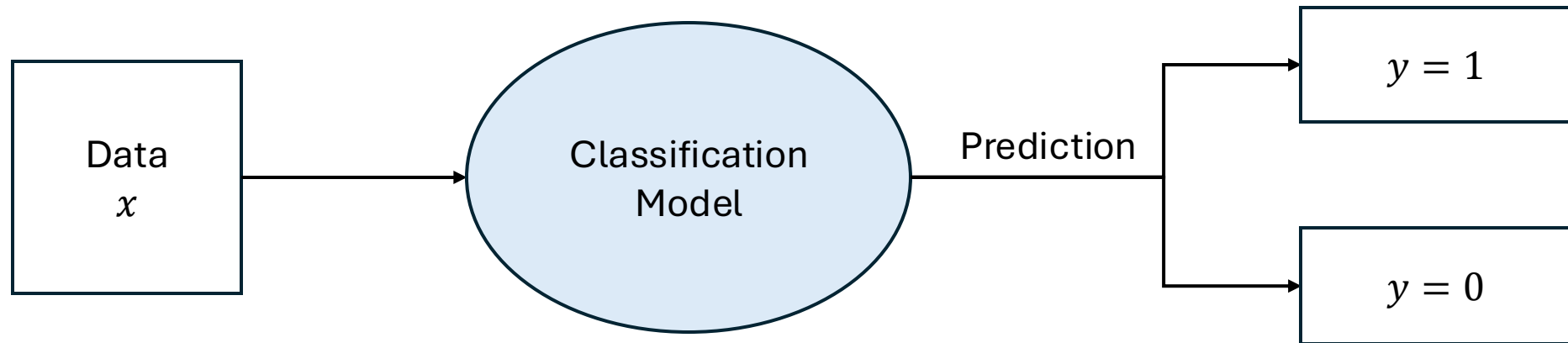
What if our outcome is *categorical*?

Example:

We can use **logistic regression** to perform classification.

Classification

- In classification, we try to predict an outcome which has *classes*:
 - Yes ($y=1$) / No ($y=0$) (binary classification)
 - Dog/Cat/Horse (multi-class classification)



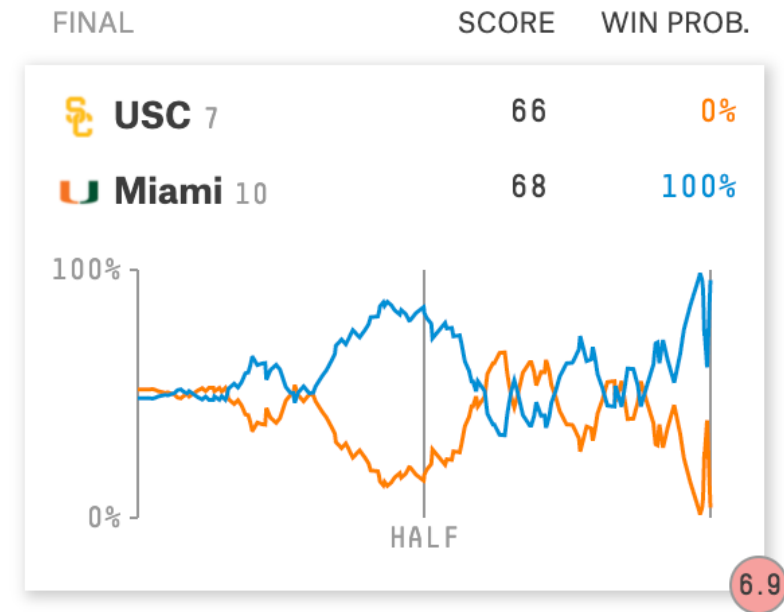
Logistic regression in the world

1. Prediction of adverse pregnancy outcomes (stillbirth, preterm birth)
2. Concussion diagnosis



Logistic regression in the world

- 3. Fraud detection in financial transactions
- 4. Spam detection in emails
- 5. Sports win probabilities



Live win probabilities

While games are being played, our interactive graphic displays a box for each one that shows updating win probabilities for both teams, as well as the score and the time remaining. These probabilities are derived using **logistic regression** analysis, which lets us plug the current state of a game into a model to produce the probability that either team will win the game.

What day is it today?

Logistic regression in the world

6. Election outcome prediction

Forbes

BREAKING

Pennsylvania Is This Year's Most Crucial Swing State—Here's How Harris Or Trump Could Still Win Without It

Mary Whitfill Roeloffs Forbes Staff

Mary Roeloffs is a Forbes breaking news reporter covering pop culture.

Follow

Why Pennsylvania could be the key to the White House

OCTOBER 21, 2024 · 5:00 AM ET

By [Destinee Adams](#), [Obad Manuel](#)

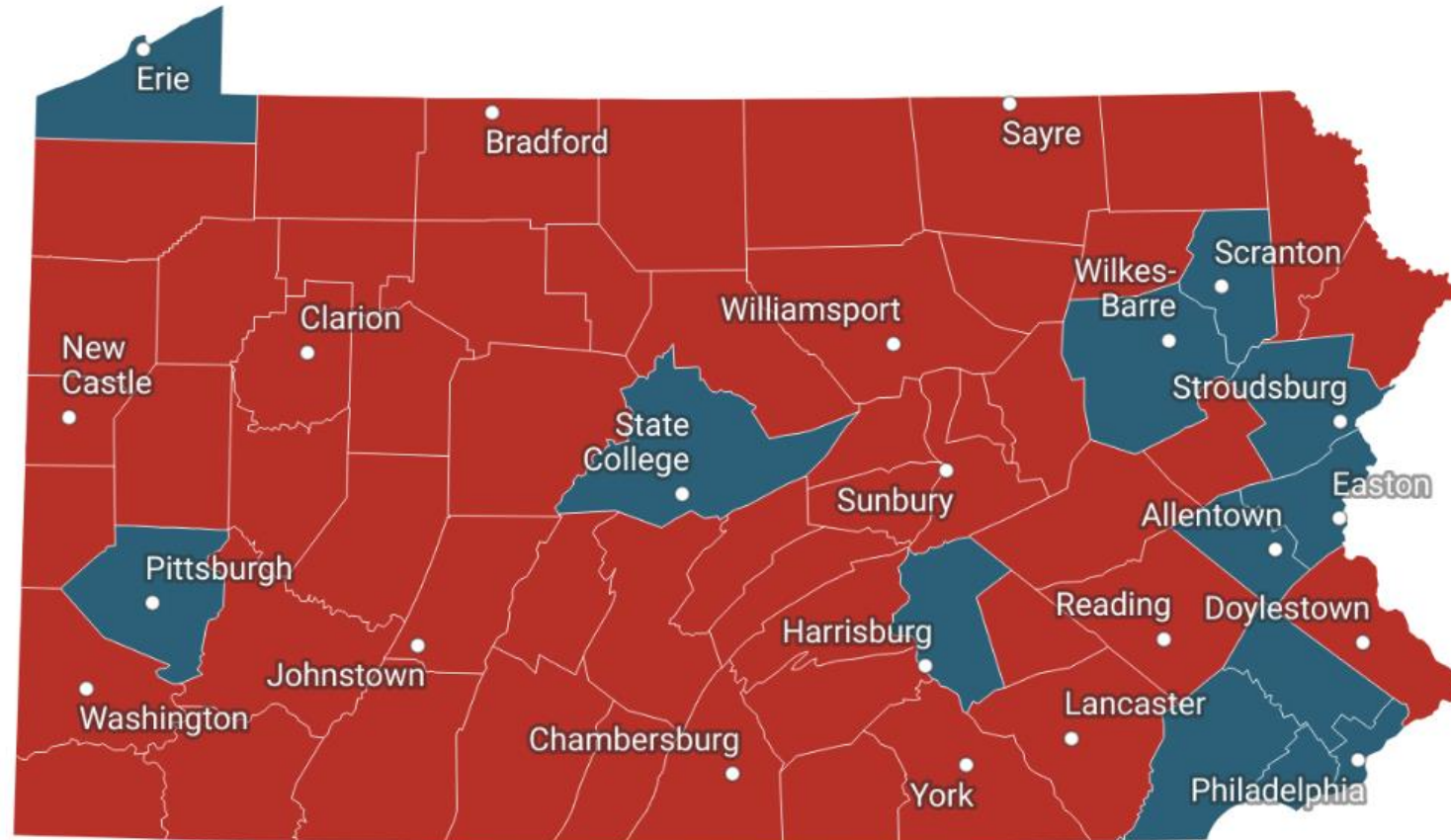
Scenario...

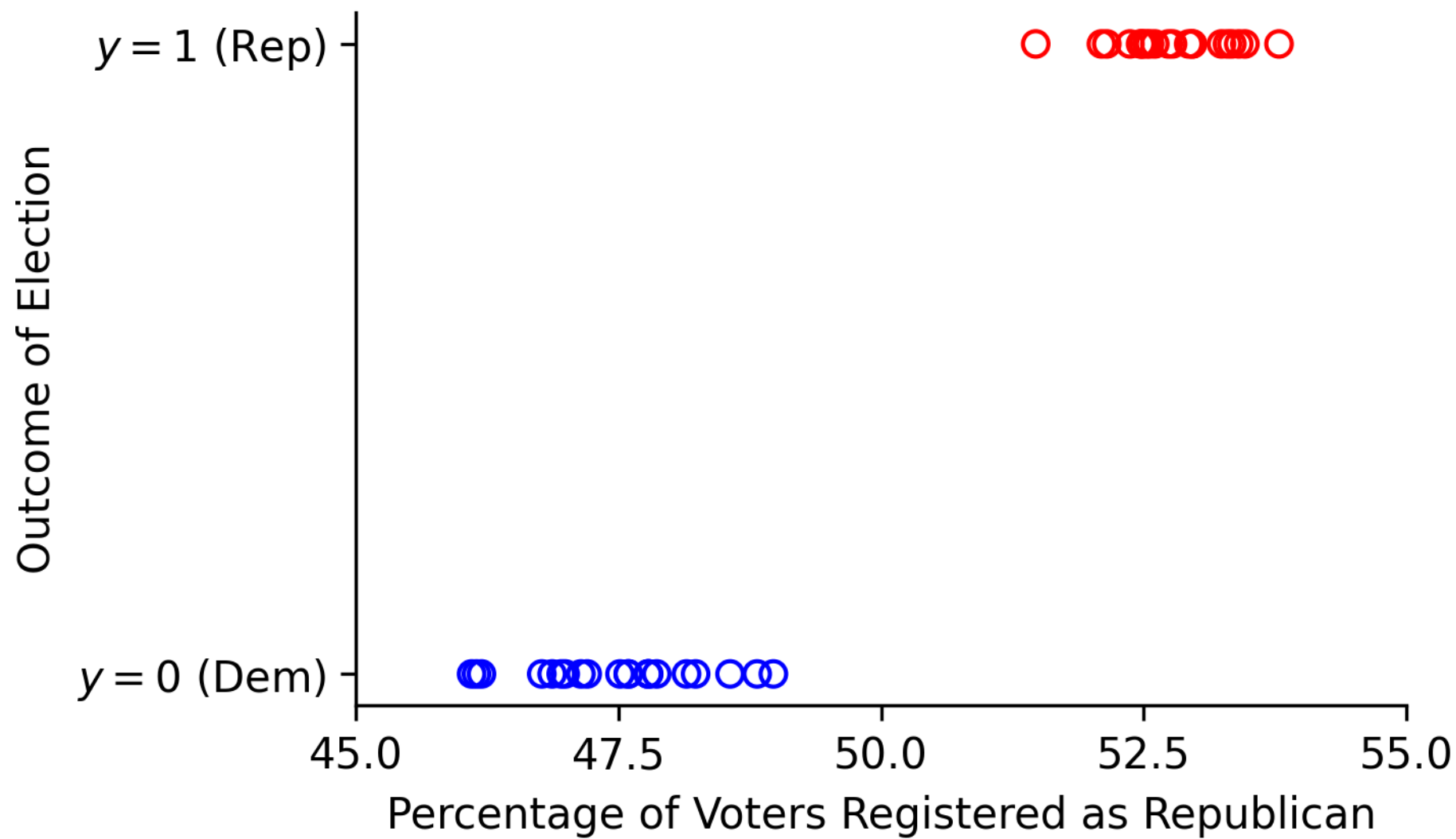


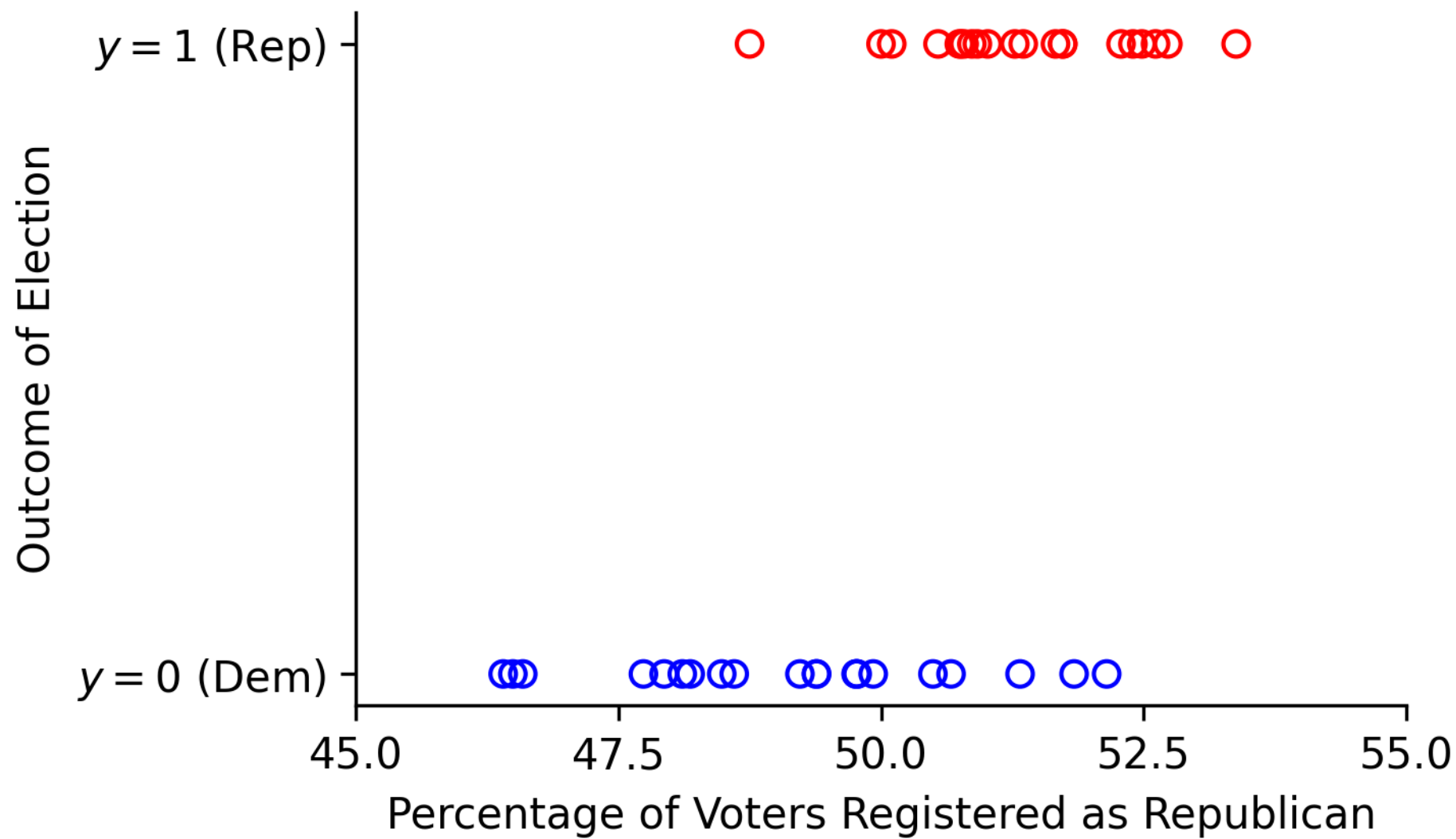
What feature(s) can we use?

Voter Registration Advantage

■ D+ ■ R+







Logistic regression model

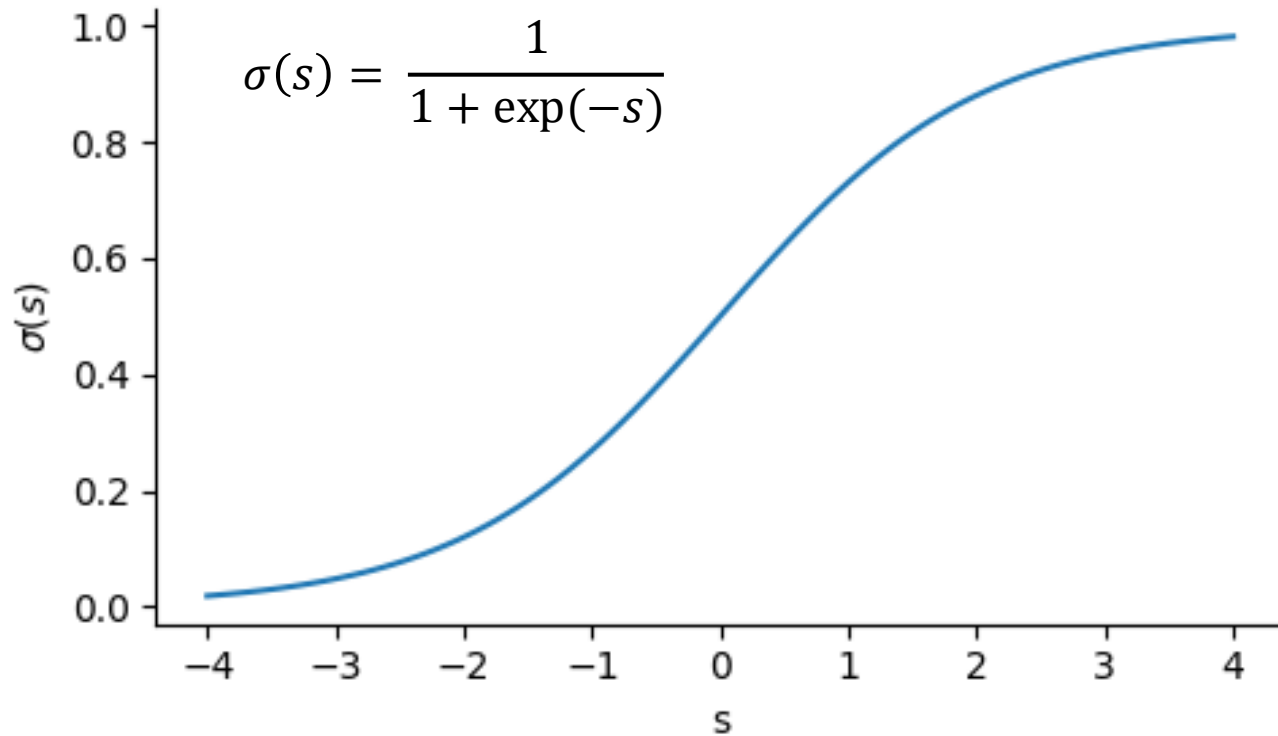
- Logistic regression models the *probability* of the outcome ($y = 1$) given the input data x .

$$f(x) = \mathbb{P}(y = 1 \mid x)$$

- Equation for logistic regression:

$$f(x) = \frac{1}{1 + e^{-(b_0 + b_1 x)}} = \frac{1}{1 + \exp(-(b_0 + b_1 x))}$$

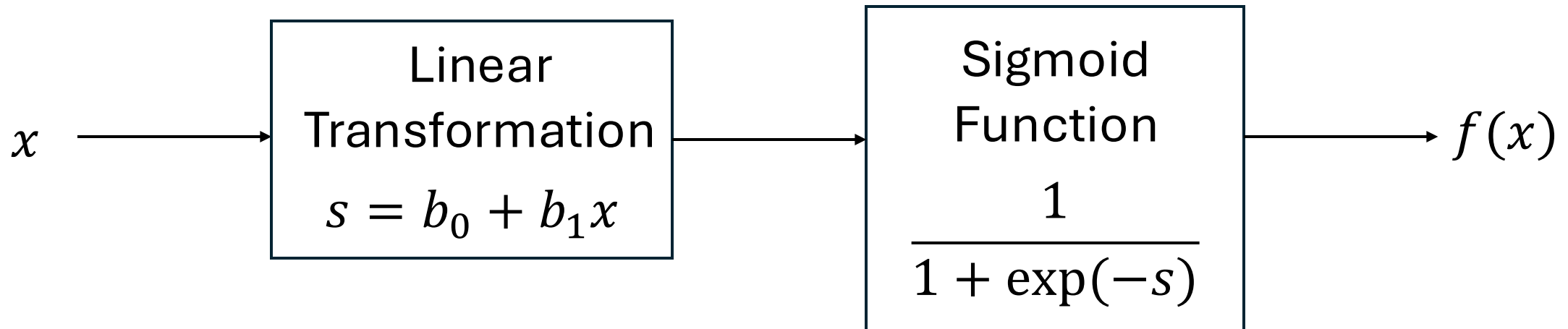
The sigmoid (logistic) function



- Domain: $(-\infty, \infty)$
- Range: $(0,1) \rightarrow$ Probability!
- Calculate:
 - $\sigma(0)$
 - $\sigma(100)$
 - $\sigma(-100)$

Logistic regression model: Another view

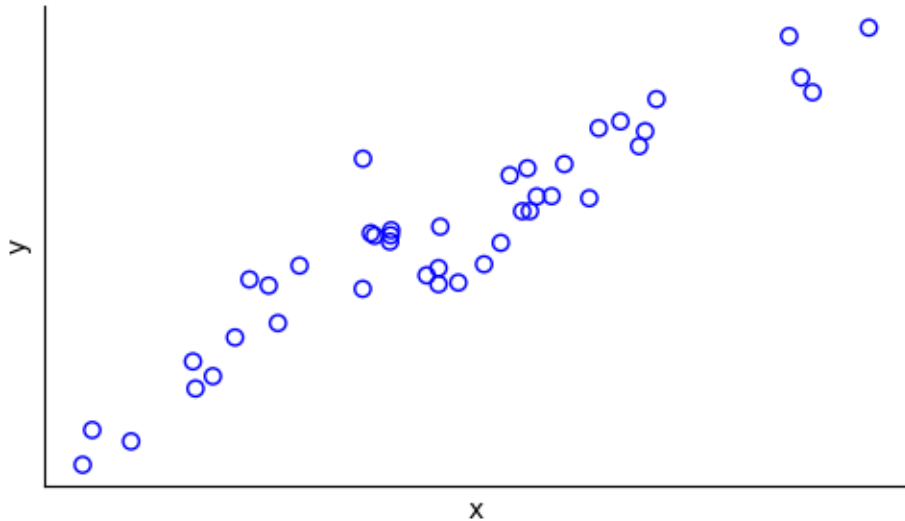
$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1x))}$$



How do we fit the model?

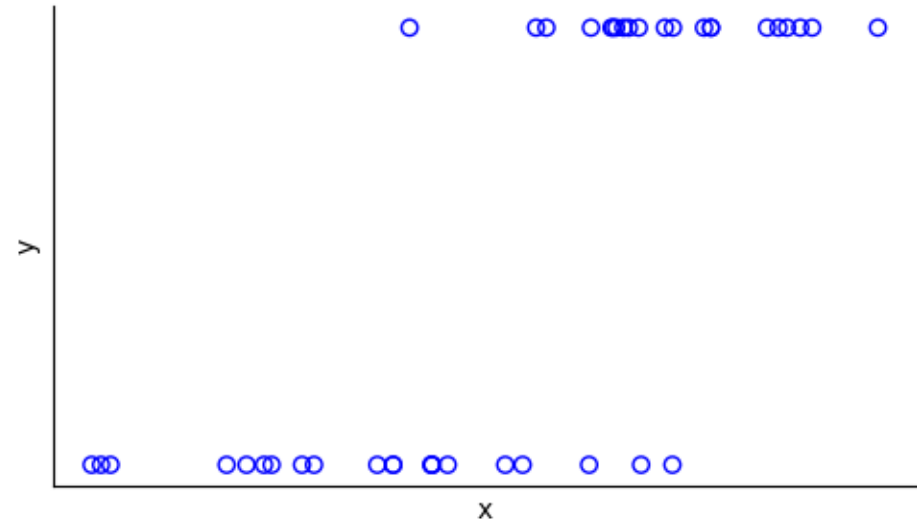
Linear regression

- $y = b_0 + b_1x$
- Find parameters b_0, b_1 for the best-fitting line:



Logistic regression

- $f(x) = \frac{1}{1 + \exp(-(b_0 + b_1x))}$
- Find parameters b_0, b_1 for the best-fitting logistic function:



Follow along in Jupyter Notebook!

Download slides/code/data: www.sunju-lee.com/demo



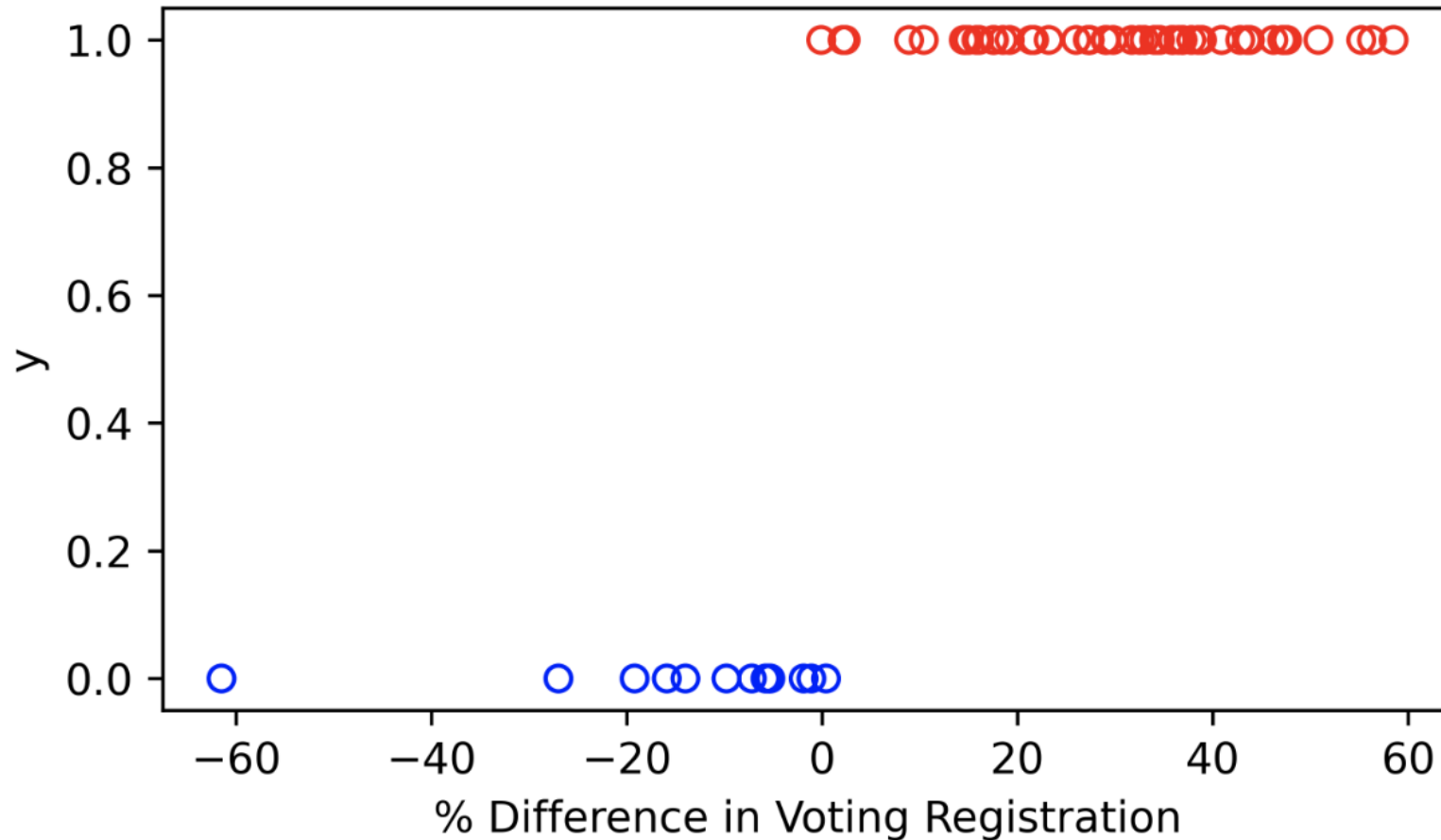
```
# What does the data look like?
```

```
pa = pd.read_csv("data_pct_diff.csv")  
pa
```

	county	percent_diff_rep	vote_2020
0	Adams	31.7	1
1	Allegheny	-27.0	0
2	Armstrong	35.8	1
3	Beaver	2.1	1
4	Bedford	56.3	1
...	...	...	...
61	Washington	10.4	1
62	Wayne	33.0	1
63	Westmoreland	14.7	1
64	Wyoming	34.6	1
65	York	21.4	1

```
# Plot the data
```

```
fig = plt.figure(figsize=(5,3))  
ax = fig.add_subplot(111)  
ax.scatter(X[y==1],y[y==1], marker = 'o', facecolors='none', edgecolors='r')  
ax.scatter(X[y==0],y[y==0], marker = 'o', facecolors='none', edgecolors='b')  
plt.xlabel("% Difference in Voting Registration")  
plt.ylabel("y")  
plt.show()
```



```
# Fit a logistic model, get the coefficient and intercept
```

```
from sklearn.linear_model import LogisticRegression  
clf = LogisticRegression(penalty = 'none').fit(X, y)
```

```
print("Coefficient: ", clf.coef_[0][0])  
print("Intercept: ", clf.intercept_[0])
```

Coefficient: 1.7065090595628383

Intercept: -0.3783588660559433

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1 x))}$$

- b_0 is the intercept
- b_1 is the coefficient

```
# Plot the logistic function on top of the data
```

```
fig = plt.figure(figsize=(5,3))  
ax = fig.add_subplot(111)
```

```
# Get xs and ys for the logistic function
```

```
xs = np.linspace(-60,60,1000)
```

```
def f(x):
```

```
    return 1/(1+np.exp(-(clf.intercept_[0] + clf.coef_[0][0]*x)))
```

```
ys = f(xs)
```

```
ax.scatter(X[y==1],y[y==1], marker = 'o', facecolors='none', edgecolors='r')
```

```
ax.scatter(X[y==0],y[y==0], marker = 'o', facecolors='none', edgecolors='b')
```

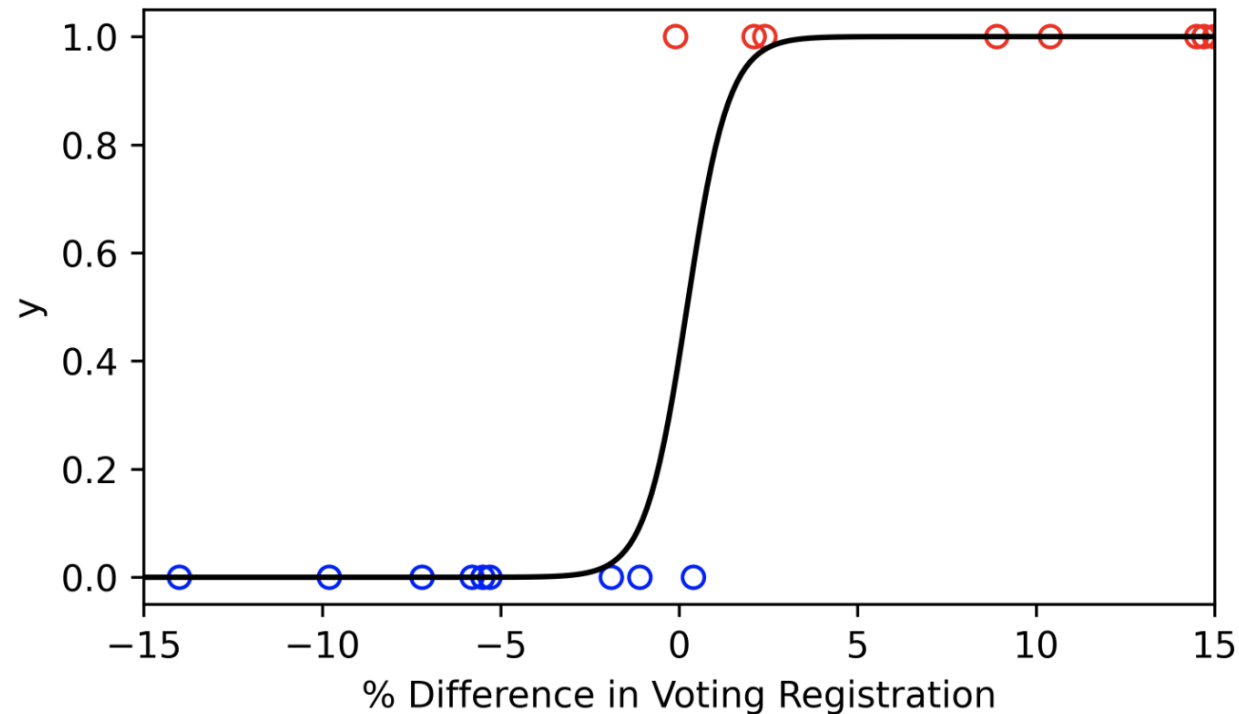
```
ax.plot(xs, ys, 'black')
```

```
plt.xlim(-15,15)
```

```
plt.xlabel("% Difference in Voting Registration")
```

```
plt.ylabel("y")
```

```
plt.show()
```



Prediction time!

Let's say that Union county's % difference in voting registration = 0.465



```
# New data
```

```
X_new = 0.465
```

```
# Get prediction – using Python
```

```
clf.predict_proba(np.array(X_new).reshape(-1,1))
```

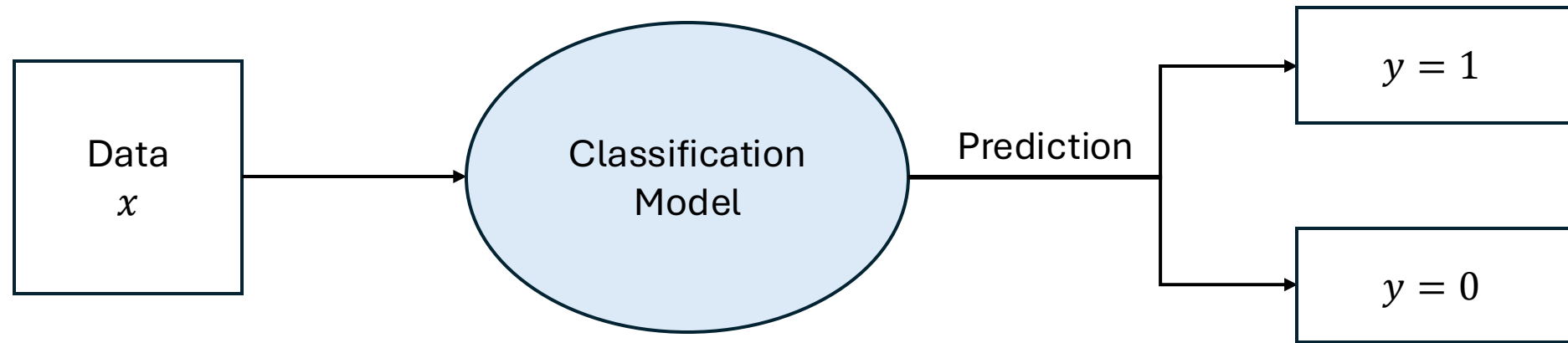
```
array([[0.39767362, 0.60232638]])
```

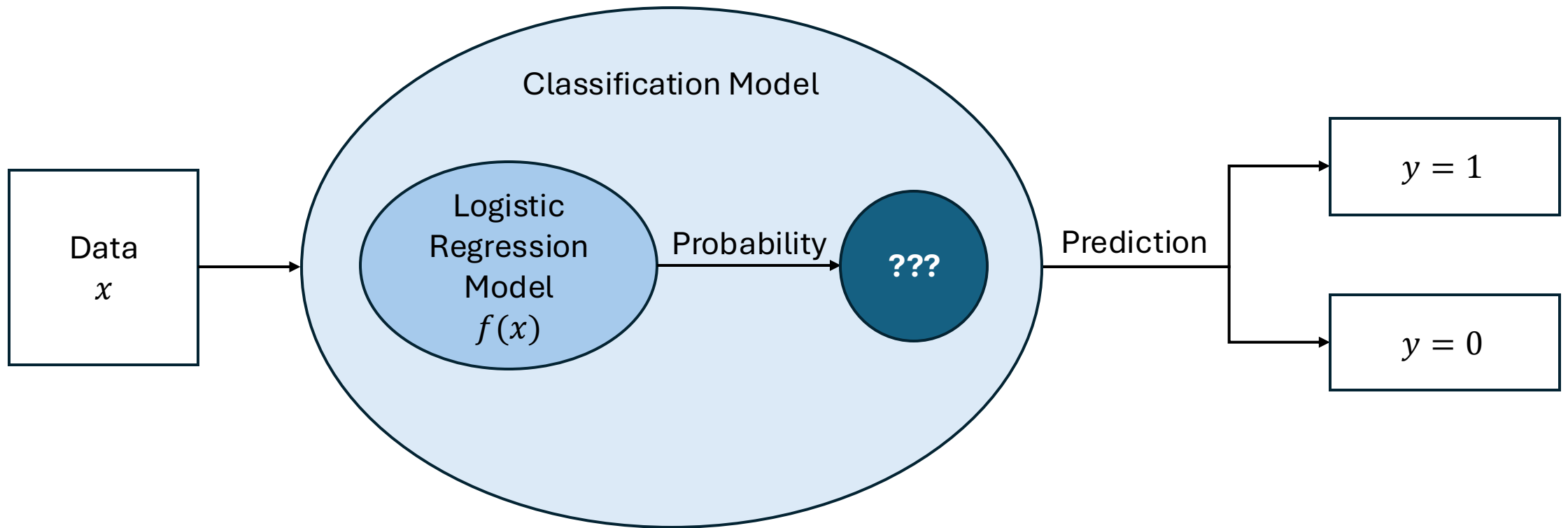
```
# Get prediction – using the logistic function directly
```

```
1/(1+np.exp(-(clf.intercept_[0] + clf.coef_[0][0]*X_new)))
```

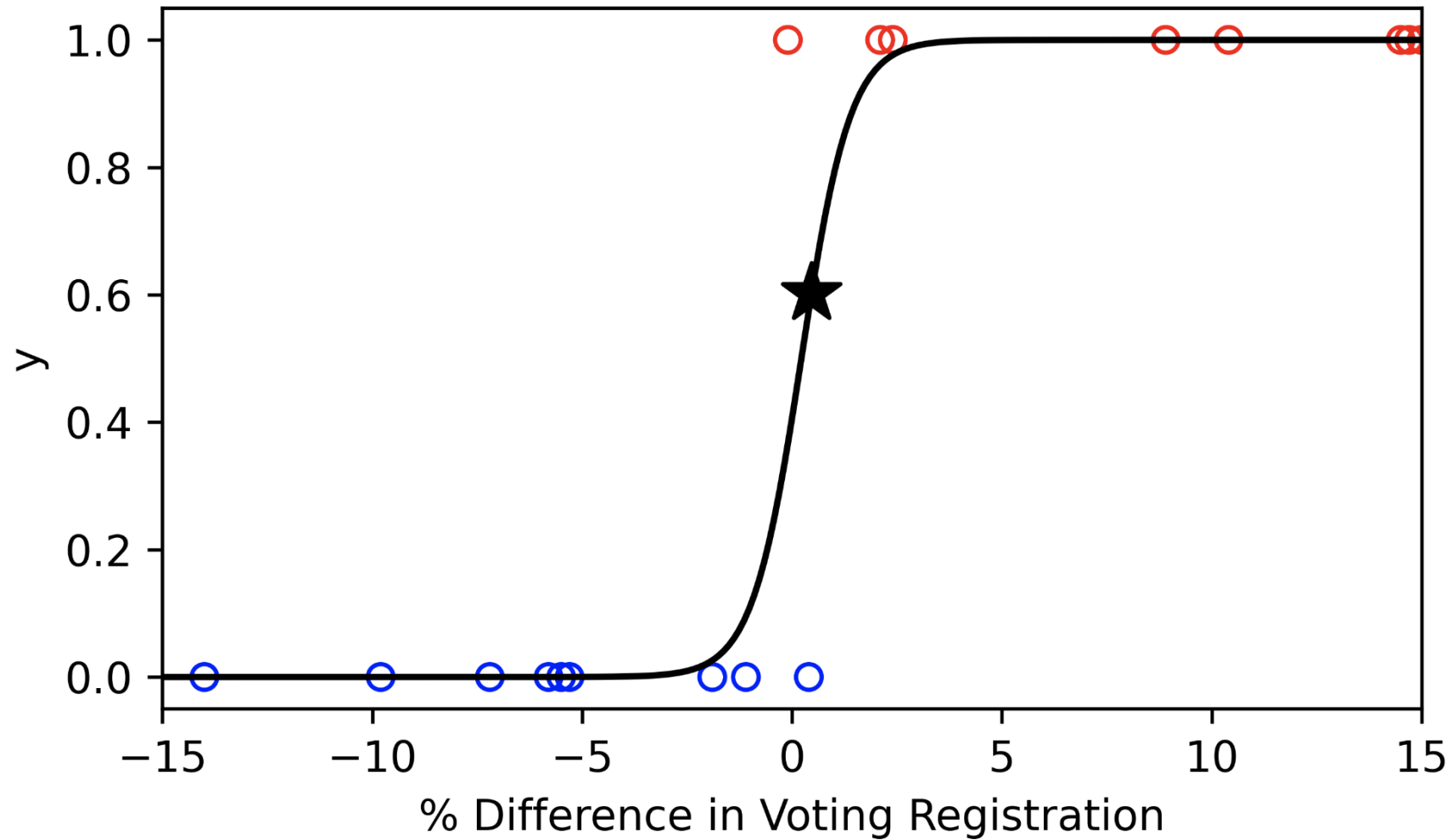
```
0.6023263817693792
```


Binary classification





How can we turn logistic regression into classification?

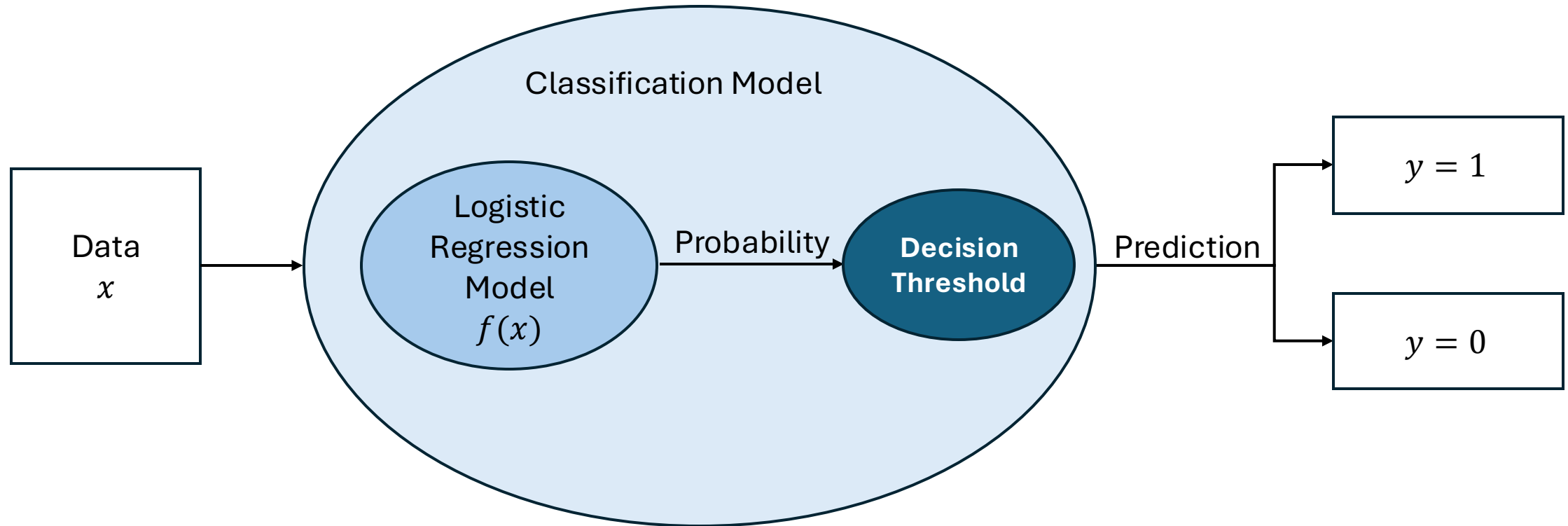


What should we predict as the election result?

What should we predict?

- Predict y based on a **decision threshold** $t = 0.5$:
 - If $f(x) \geq 0.5$, predict $y = 1$
 - If $f(x) < 0.5$, predict $y = 0$
- Note:
 1. The *decision boundary* is linear!
 2. Depending on the problem, other values of t may be better

Binary classification using logistic regression



Preview of more topics in logistic regression

We focused on one predictor variable, but there can be more...

Simple logistic regression

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1x))}$$

Multiple logistic regression

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1x_1 + b_2x_2 + \cdots + b_{100}x_{100}))}$$

Preview of more topics in logistic regression

We focused on binary classification, but we can do multi-class classification...

Simple logistic regression

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1 x))}$$

(sigmoid)

Multinomial logistic regression

$$f_k(x) = \frac{\exp(b_0^k + b_1^k x)}{\sum_{j=1}^K \exp(b_0^j + b_1^j x)}$$

(softmax)

A quick
aside...

2024 Nobel Prize

2024 Nobel Prize in physics
awarded to John J. Hopfield,
Geoffrey E. Hinton for
discoveries that 'enable
machine learning with artificial
neural networks'



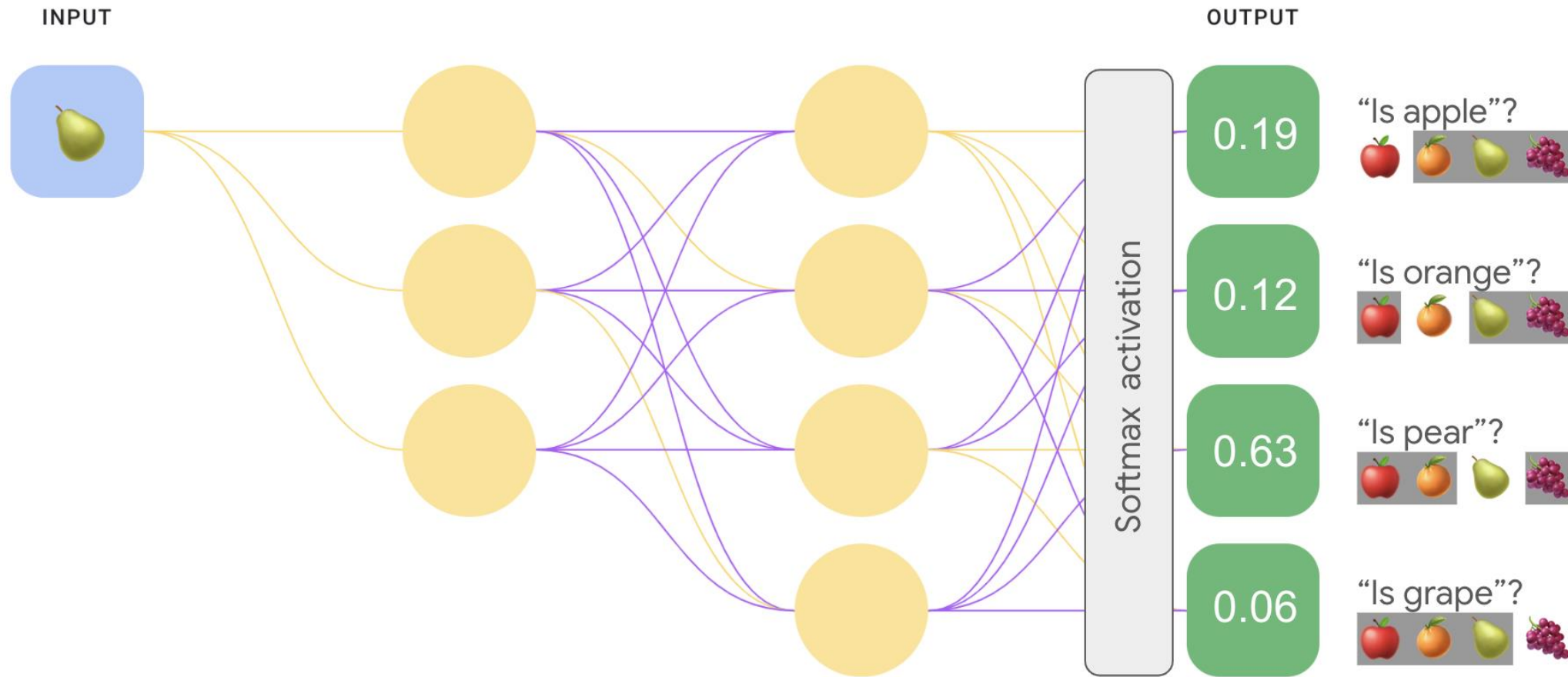
John J.
Hopfield



Geoffrey E.
Hinton



Neural networks often use the softmax function!



Just like sigmoid: transforms numbers into class probabilities

Preview of more topics in logistic regression

Logistic regression is an *interpretable* model: we can interpret the coefficients.



Bernard Parker, left, was rated high risk; Dylan Pugett was rated low risk. (Josh Ritchie for ProPublica)

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016

How We Analyzed the COMPAS Recidivism Algorithm

by Jeff Larson, Surya Mattu, Lauren Kirchner and Julia Angwin
May 23, 2016

Risk of General Recidivism **Logistic Model**
Dependent variable:
Score (Low vs Medium and High)

Female	0.221*** (0.080)
Age: Greater than 45	-1.356*** (0.099)
Age: Less than 25	1.308*** (0.076)
Black	0.477*** (0.069)
Asian	-0.254 (0.478)
Hispanic	-0.428*** (0.128)
Native American	1.394* (0.766)
Other	-0.826*** (0.162)
Number of Priors	0.269*** (0.011)
Misdemeanor	-0.311*** (0.067)
Two year Recidivism	0.686*** (0.064)

We used those factors to model the odds of getting a higher COMPAS score. According to [Northpointe's practitioners guide](#), COMPAS "scores in the medium and high range garner more interest from supervision agencies than low scores, as a low score would suggest there is little risk of general recidivism," so we considered scores any higher than "low" to indicate a risk of recidivism.

Our **logistic** model found that the most predictive factor of a higher risk score was age. Defendants younger than 25 years old were 2.5 times as likely to get a higher score

Today, we learned...

- What is logistic regression used for?
- How do logistic regression models work?
- How can we use logistic regression for classification?

What will you predict?

